

Discrete Lattice Regularization of the Cosmological Constant and Chiral Topology in $T^7 \rightarrow M^4$ Spacetime Projections

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Abstract

We formalize a discrete, non-singular block universe framework where 4-dimensional Minkowski spacetime (M^4) acts as a projection layer wrapping into a higher-dimensional 7-toroidal manifold (T^7) through a Möbius boundary condition. By enforcing a 6×6 spatial-imaginary matrix whose eigenvalues yield an exact trace-free condition ($\text{Tr}(M_6) = 0$), high-energy zero-point bulk fluctuations cancel out at the Planck scale. Integrating a rationalized lattice value of $\pi_{\text{lattice}} = 22/7$ over a discrete 4D hyperspherical volume yields an exact suppression factor matching observed dark energy density without fine-tuning. Furthermore, this discrete architecture structurally resolves the thermodynamic arrow of time, time-reversal asymmetry, and left-handed biological homochirality as direct geometric consequences of the macroscopic Möbius topology and higher-dimensional clock synchronization.

1 Preface: The Cosmic Engine and the Voxel Brane

Picture the universe not as a smooth, empty void, but as a colossal, high-dimensional engine. Imagine a vast, invisible dynamo—a seven-dimensional toroidal condenser (T^7)—spinning continuously along a seventh axis, its rhythmic pulses synchronized by an eighth dimension of time acting as a master pendulum. Wrapped around this spinning dynamo is a ribbon of reality, biting its own tail like the mythical Ouroboros. But this ribbon does not connect perfectly; it is joined with a single, subtle half-twist, forming a four-dimensional Möbius loop.

The reality we experience every day—the Minkowski spacetime we move through and the time we march along—is projected onto this twisted ribbon. If you were to zoom in infinitely close, past atoms and the smallest subatomic particles, you would not find a smooth continuum. You would find a rigid, discrete matrix. Spacetime itself is a four-dimensional “voxel brane,” a grid composed of finite, Planck-sized blocks.

This grid operates as a macroscopic cellular automaton, larger than life but functioning much like Conway’s *Game of Life*. It flickers and updates frame by frame, driven by the mechanical rotation of the toroidal condenser beneath it. We propose that everything we observe in nature is the unavoidable result of this topological matrix. Because the macroscopic ribbon has a twist, the universe possesses an inherent topological torque, dictating the left-handed chirality preference of all biological life. Because the matrix updates algorithmically, thermodynamic energy flows strictly in one direction. In this framework, gravity and electromagnetism are not separate, mysterious forces patched together; they are the natural geometric consequences of a discrete, twisted game board projecting off a 7-dimensional engine.

The mathematics that follow decode the exact operating rules of this matrix [4, 5].

2 Lattice Rationalization of π

To eliminate continuous transcendental artifacts at the Planck boundary, spatial coordinates are quantized across a rationalized lattice stencil. The lattice-derived value of π is defined by the 12-point anchor matrix normalization over orthogonal voxel boundaries:

$$\pi_{lattice} = \frac{22}{7} = 3.\overline{142857}$$

The voxel grid discretization factor δ_π governing spatial closure is expressed as:

$$\delta_\pi = \left| \pi - \frac{22}{7} \right| \approx 1.264 \times 10^{-3}$$

3 The 6×6 Spatial-Imaginary Matrix Eigenvalues (M_6)

Internal phase propagation and vacuum regularization are governed by the 6×6 spatial-imaginary matrix M_6 . The trace-free condition ensuring bulk zero-point energy cancellation ($Tr(M_6) = 0$) is defined by conjugate eigenvalue pairs λ_i :

$$M_6 = \begin{pmatrix} 0 & -\omega_1 & 0 & 0 & i\tau_1 & 0 \\ \omega_1 & 0 & -\omega_2 & 0 & 0 & i\tau_2 \\ 0 & \omega_2 & 0 & -\omega_2 & 0 & 0 \\ 0 & 0 & \omega_2 & -\omega_3 & 0 & 0 \\ -i\tau_1 & 0 & 0 & i\tau_1 & -\omega_4 & 0 \\ 0 & -i\tau_2 & 0 & 0 & \omega_4 & -\omega_4 \end{pmatrix}$$

The characteristic equation determining allowed discrete energy and momentum eigenmodes on the lattice yields:

$$\det(M_6 - \lambda I_6) = \sum_{n=0}^6 c_{2n} \lambda^{2n} = 0$$

where the trace-free constraint strictly enforces $c_{10} = Tr(M_6) = 0$ eliminating ultraviolet zero-point divergences.

4 Discretized Hyperspherical Volume & Correction (ΔV)

Replacing continuous transcendental geometry with discrete voxel scaling on the 4D Minkowski hypersphere bounded by the Möbius loop yields the corrected volume element:

$$V_{4,lattice} = \frac{242}{49} R_U^4$$

where $R_U = l_v e^{1/\alpha}$ is the universal Ouroboros radius, l_v is the Planck voxel length, and α is the fine-structure constant. The discrete volumetric correction term ΔV is derived as:

$$\Delta V = |V_{4,continuous} - V_{4,lattice}| = \pi^2 R_U^4 - \frac{242}{49} R_U^4 \approx 0.00397 \cdot R_U^4$$

5 Cosmological Constant Suppression Ratio

Combining the 4D discrete volume correction with the exponential Casimir boundary scaling yields the exact vacuum energy density ratio [1]:

$$\frac{\rho_\Lambda}{\rho_{Planck}} = \frac{\Delta V \cdot \exp\left(-\frac{2}{\alpha}\right)}{Tr(M_6^2)} \approx 10^{-119}$$

6 The Planck Floor and the Thermodynamic Arrow of Time

Standard physics relies on continuous calculus, which breaks down into unphysical singularities at the Planck length. By strictly enforcing the spatial parameters of our M^4 matrix to the discrete 4×4 Yee-lattice [2], the Planck length ceases to be an arbitrary lower bound; it is the fundamental resolution limit of spacetime geometry.

Furthermore, this discrete architecture resolves the paradox of time-reversal symmetry [3]. While microscopic field equations operate reversibly in spatial coordinates, macroscopic time strictly moves forward due to increasing entropy. In our $T^7 \rightarrow M^4$ framework, time is not a traversable spatial dimension. Instead, time evolution is driven by the rotation of the 7th-axis ($dt > 0$), acting as an external algorithmic clock update. The state vector $\Psi(x, t_7)$ updates at the Planck frequency ($\nu_{Pl} = c/l_v$). Because the 7th-axis clock unconditionally drives the computational progression of the voxel grid, macroscopic T-symmetry is structurally broken, enforcing a strict unidirectional thermodynamic arrow of time.

7 Topological Enforcement of Biological Homochirality

The $T^7 \rightarrow M^4$ framework provides a geometric resolution to the established preference for L-amino acids in biological structures [6]. Because the M^4 projection is wrapped with a single Möbius half-twist ($N = 1$), the spacetime boundary is non-orientable. This introduces a persistent left-handed topological torque across the macroscopic universe.

As molecular chains self-assemble in the early prebiotic environment, left-handed conformations (propagating “with the flow” of the native Möbius twist) encounter lower lattice impedance across the discrete M_6 spatial stencils than right-handed configurations. Over macroscopic time scales, this lower energetic stabilization barrier fundamentally phase-locks biochemical evolution, dictating that life utilizes L-amino acids not by random accident, but as a resonance alignment with the macroscopic chiral geometry of spacetime.

8 8th-Axis Synchronization and Topological Phase-Gain

The thermodynamic arrow of time is stabilized by a higher-dimensional coupling. The 7th-axis update tick (ω_7) is phase-locked to a sweeping master envelope driven by the 8th-axis (θ_8). The interaction between the local computational tick and the macroscopic cosmological envelope is governed by the coupled oscillator equation:

$$\frac{d^2\Psi}{dt_7^2} + \omega_7^2(1 - \Delta V \cos(\theta_8))\Psi = 0$$

Because the physical universe is an open thermodynamic system relative to the 7D bulk, a cyclical quantum system (a chiral wave packet) completing a full 2π circuit along the non-orientable Möbius loop does not return to its initial energetic state [7]. The topological phase-gain integrated over one complete cycle yields:

$$\Delta\Phi_{net} = 4\pi^2 \left(\frac{\Delta V}{\alpha} \right) \cdot \exp \left(-\frac{1}{\alpha} \right)$$

This coherent phase-shift allows the universe to constantly draw rotational maintenance energy from the bulk without violating local conservation laws, maintaining the rigidity of fundamental constants across deep time.

References

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